

Data Analytics - Research Assignment 1

SECTION A : Data Fundamentals

1) According to my understanding, a database is a place where data that was collected in real time, is stored and organized in a way that makes it easy to retrieve later.

I believe databases are essential in the modern world because it's mostly digital and fast-paced, therefore they help keep large amounts of important information efficiently.

Examples :

→ The Spotify music app, whose database can collect and keep a user's listening patterns, from the most listened to song or artist and recommend similar artists or genres for the user later.

→ A bank can track a user's spending patterns with each transaction, categorizing it into groceries, entertainment, etc, and keeping it, while also allowing the user to keep track of their habits on the banking app.

2) A datawarehouse is a centralized system where cleaned historical data is stored, and unlike a regular database which keeps record of live transactions, a datawarehouse stores that has been organized and cleaned.

I think a company would use a datawarehouse to store all its data taken from all its aspects and putting them in one place.

③ A Data Lake is a repository that stores large amounts of data in its original format, whether structured, semi-structured or unstructured all on the same platform. According to my understanding, a data lake stores data in its raw form, whereas a datawarehouse houses data that is organized or structured.

- According to an article by Velocity Insights, data lakes are more flexible, and can handle large volumes of data at a lower cost.

I believe businesses would prefer data lakes as a cost-effective solution to housing diverse data and accessing it when they need to.

④ From what I can understand, a Data Mart is a smaller version of a datawarehouse, and is usually focused on a specific analysis of data. I believe it relates to a datawarehouse in the sense that, they both carry the same objective, but with a data mart being on a smaller scale. Typically datamarts would be used by companies, but for data relating to maybe their sales or marketing department.

⑤ Structured Data is data that follows a pre-defined format, and is usually in a table that contains rows and columns, making it easy to analyze.

Example - a class register that is in table format, with columns specifying name, surname, age and signature, and rows to fill in the information

Semi-structured Data is data that is not stored in tables, or neat table and rows, but uses tags and/or labels to keep it organized and somewhat easy to analyze

Example - My phone's gallery, where images and videos are organized into labelled folders e.g camera, snapchat etc, making it easy for to know where to look.

Unstructured Data - unlike the above mentioned, unstructured data is free flowing, is not predefined into spreadsheets and contains no tags or tables.

Example - an image or video which require a person to look or watch, analyze and organize it into either semi-structured data, or create structured using data from the analysis.

6) Data Modelling is described as the process of creating a structured and visual representation of data and defines how data elements relate to one another in the same system. I think it is important when designing a system because it assembles the needs of a business and turns them into accurate organized data that can influence decision-making

Examples :

Conceptual Data whose function is to give a big picture view of data, what data is contained and its condition.

Logical Data Model whose function is to map conceptual data into technical data structures, with more details about data concepts & relations identified in the conceptual model.

7) According to my understanding, data mining is a process that already exists is analysed for any valuable informations or patterns that may have been missed. Data mining can be used by businesses to find knowledge that can be used to increase to increase customer trust and new revenue or regain customers.

Examples

- In education, DM can help to study algorithms to test students, customize lessons and help educators see what support students need.

- Data Mining can also be used by manufacturing companies to analyse their historical data to predict wear of machinery and anticipate issues, allowing for better maintenance scheduling and increasing efficiency.

② A database management system (DBMS) is a platform that an data analyst would use to retrieve data from a warehouse to analyse.

Examples of DBMSs are:

1) MySQL, whose aim is to store, organize and retrieve data, and I believe this system is suitable for analysing structured data

2) PostgreSQL which is a system that is known for its reliability flexibility and support. I believe this system can be used by businesses that require accurate data collection at a low cost.

3) Snowflake is a DBMS built primarily for online analytical processing and data warehousing on a large scale. I believe this system would be suitable for housing all data types to store, use and share, whether structured, semi or unstructured

4) Lastly, Oracle DBMS that stores, organizes and retrieves like the others, but focuses on Online Transactional Processing (OLTP). I believe it would be suitable for the retrieval of data collected through OLTP which happen in real time.

④ ETL is an acronym for Extract, Transform and Load, and is the process used to clean and organize data before it can be stored in data warehouses. Extract refers to the process of retrieving data from a database, Transform being the sorting of data into structures, and Load being the transferring of data onto a warehouse

⑤ According to my understanding, a data format is a standard pre-defined structure for loading data onto a file

Common format include:

→ CSV format, which is a simple text-based data format in table form with each line of data being represent by a column and field in a row separated by commas. This format can be used to easily share data across different systems

→ JSON is data format that is also text-based and can be used for the storage and sharing of data that can be easily read by both humans and computers. This format can be used to transfer data in between applications.

→ Parquet is another data format, usually column based and is used for managing large sets of data and real-time big data processing.

I think this format would be suitable in the analysis of social media data.

SECTION B : Artificial Intelligence & Machine Learning

11) Artificial Intelligence is technology that allows computers to do tasks that would be normally be done by humans.

Narrow AI is AI that is programmed to perform only one task at a time, and an example of this AI would be asking Google Gemini to open an app.

Generative AI, also known as strong AI is a theory and doesn't actually exist but is a hypothetical form of AI that would be able to understand, learn and perform tasks like a human. Eg. a robot that can drive.

12) Machine Learning is an element of AI that enables computers to learn patterns from data and improve their tasks without being explicitly programmed to.

Types of ML:

→ Unsupervised Learning which means the AI learns from unlabelled or unstructured data, meaning it must attempt to find patterns or groups on its own.

→ Supervised Learning enables the AI to learn from data that is already labelled or structured.

→ Reinforcement Learning is a type of ML that allow the AI to learn to perform tasks without guidance, meaning it must learn from interacting with its surroundings

13) Deep Learning is a method of AI that teaches computers to process information in a way that a human would. I believe that Machine Learning, AI and deep learning are interlinked with AI being the one enabling computers to learn/perform tasks, ML enabling computers to recognise and learn patterns and DL teaching computers to perform task that require human intelligence.

14) Natural Language Processing is a technology that allows computers to understand, translate and manipulate human language.

Examples of NLP applications are:

- Google Gemini
- Auto-correct
- Meta AI

15) Large Language Model is a type of machine learning that has been trained on large amounts of data, allowing to understand and respond like a human would. In comparison to standard ML, LLMs are designed to deal with different tasks at once.

We know Large Language Models are:

- 1) Generative LLM's whose function is to generate new text-based on what they received
- 2) Encoder based LLM which are designed to understand and analyse instead of generating long texts

16) Generative AI is AI that has been trained to create new content, and ideas, learn the human language and reuse its knowledge to solve new problems. On a high level it acts as a model that generates

what you asked for.

3 tools that use Generative AI are

- ChatGPT - which works with text and helps answer questions, write, summarise and brainstorm.
- Adobe Firefly - which helps with image generation and editing when asked to
- Grammarly - helps rewrite, improve, summarise and generate texts.

17) Training data is the information used to teach machine learning models how to recognize patterns and generate content. In order to produce results that are reliable, it is important that the AI is fed data that is accurate, relevant or complete. Since AI learns from the data it is given, I think it is important it's important that the training data is not incorrect or biased, so that it does not produce inaccurate results.

18) Computer Vision is AI that enables computers with the ability to process, analyze and interpret visual data such as pictures and videos.

Example:

- Google image search, which allows you to attach an image and let the AI produces results based on what it can see on the picture.
- Facial recognition which we use mostly on phones as a security measure, studies and keeps our facial features and recognizes us to unlock the phone

19) Application Programming Interface (API) is a mechanism that enables two software components to communicate with each other. A real world analogy I can think of is when making an online payment and a request is sent after I am redirected to a portal, that send a message to my bank for me to approve payment request.

I believe API's are important because they allow different companies to communicate and share data/services without having to create their own platforms for those services.

20) API keys are the one used to verify the programme and their aim is to identify and authenticate any application that wants to access an API. API keys should always be protected and are commonly stored in environment variables, secret management systems or secure configuration systems. I believe that if exposed publicly, a number of people would gain access to APIs without permission.

21) REST is an acronym for Representational State Transfer and is said to be a definitive set of functions that clients can use to access server data using HTTP.

The main methods used in HTTP are:

- GET - whose function is to retrieve data
- POST - whose function is to create or send new data
- PUT - whose function is to update or replace existing data
- DELETE - whose function is to remove data

22) JSON is an acronym for JavaScript Object Notation, and is a text-based format that is used to store and exchange structured data, and is commonly used in API as it can be read by both humans and computers

Example

```
{ "name": "karabo",  
  "age" : 24,  
  "city" : "cape town"  
}
```

SECTION D: Data Ethics, Governance and Careers

29) Data Privacy is a policy or law put in place, allowing people the control over their personal data and the right to choose how this data can be collected, stored and used by organizations.

I think this policy is important as it ensure the people's personal data is protected from theft and being used without permission.

- The General Data Protection Regulation is a law around Privacy and security that focuses on lawfulness, purpose limitation, minimization and accuracy to name a few. This law was passed by the European Union in 2018, and applies to all countries that are members of the EU as well as countries within the European Economic Area, although it may also apply to are outside Europe but serve those countries.

30) Data Ethics is ethics that focuses on the studying and moral implications that may come with data collection.

Examples of unethical data use

- companies that collect personal information such as emails & phone numbers and selling it to other companies, which is problematic as the owners did not consent
- creating images using AI to advertize and gain customers, which is problematic because it is false advertizing.
- taking photos of people from the internet and using them to generate pornographic, which may be harmful as it violates privacy and may cause reputational damage

31) Data Governance is a framework that includes policies and roles that an organization would use to manage its data properly.

key components that a good data governance framework should include:

- 1) Data Quality which ensures that data is accurate, complete, consistent and reliable
- 2) Security which protects data from being accessed by unauthorized persons, lost or stolen
- 3) Roles and Responsibilities which defines who's responsibility it is to manage and protect data
- 4) Policies and Standards which creates rules for how data should be collected, stored, shared and used.
- 5) Compliance which ensures that organizations follow the relevant rules and regulations such as the GDPR.

32) Because AI was created by humans, it means it can also be biased based on the data that it was trained on. A biased AI model would mean that it produces results favours/disadvantage certain demographics.

For example a banking system that uses AI that has been taught to favour credit applications to a certain race over the other, which may be problematic as it may imply the bank is racially prejudiced.

83)	ROLES	SKILLS	TOOLS
	Data Analyst	- analyses historical data to find patterns or trends	- Excel - MySQL
	Data Scientist	- use historical data and stats to predict and solve problems in the future	- Python, SQL, Spark, Airflow
	Data Engineer	- creates and maintains the systems that collect, store and process data	Python, SQL, AWS, Spark
	AI Engineer	builds, integrates and deploys AI and machine learning applications	Python, Tensorflow, APIs, cloud platforms

34) Business Intelligence (BI) is when data is used by businesses, through technological tools to help them make informed business decisions.

Examples of BI are

→ Microsoft Power BI, which turns raw data into visual data such as reports and charts.

→ Tableau, like Microsoft also transforms raw data to create visual representations.

Both these BI tools can help businesses see trends and patterns, giving them the chance to monitor and compare in order to make informed decisions.

SECTION E : Personal Reflection

35) The most surprising concepts for me were Natural Language Processing and Computer Vision, and I found these interesting because I had been interacting with for years and never would have known that they form part of AI or that they are actual concepts with definitions.

36) A concept I struggled with was the Machine Learning concept and the aspects that form part of it. I used articles and YouTube to try and understand a bit better.

37) Career-wise, Data and AI would possibly eliminate my role as it can be automated and order capturing and billing can be computerized. In everyday life AI would be beneficial, especially when it comes to accessing information, and decreasing the amount of time it would take to write an essay for instance.